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ORIGINAL PAPER

Plasma tumor necrosis factor-α and C-reactive protein as biomarker for survival in head and neck squamous cell carcinoma

Bengt-Åke Andersson · Freddi Lewin · Jan Lundgren · Mats Nilsson · Lars-Erik Rutqvist · Sture Löfgren · Nongnit Laytragoon-Lewin

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Abstract
Purpose Tumor TNM staging is the main basis for prognosis and treatment decision for head and neck squamous cell carcinoma (HNSCC) despite significant heterogeneity in terms of outcome among patients with the same clinical stage. In this study, a possible role of plasma interleukin-2 (IL-2), interleukin-6 (IL-6), granulocyte–macrophage colony-stimulating factor (GM-CSF), tumor necrosis factor-α (TNF-α) and C-reactive protein (CRP) as biomarkers for survival of HNSCC patients was investigated.
Methods In this prospective study, plasma levels of IL-2, IL-6, GM-CSF, TNF-α and CRP in patients ($n = 100$) and controls ($n = 48$) were analyzed.
Results Significantly elevated levels of CRP and TNF-α ($p < 0.001$) were found in the patients. Combination of upregulated CRP and TNF-α in the patient plasma was significantly related to shorter patient survival, independent of clinical stage.
Conclusions Our findings indicate that CRP and TNF-α might be suitable as biomarkers in combination with tumor TNM staging for predicting survival and individualized treatment of HNSCC patients. Plasma CRP and TNF-α analysis are simple, rapid, cost effective and suitable for clinical practice.

Keywords Head and neck squamous cell carcinoma · Biomarkers · Survival · CRP · TNF-α

Introduction

Tumor TNM staging continues to be the basis for clinical decision-making for head and neck squamous cell carcinoma (HNSCC), despite significant heterogeneity in outcome among patients with the same clinical stage. Improved diagnostic and prognostic methods are required for individualized treatment of this patient group, for which survival time only has marginally improved during the last 30 years (Forastiere et al. 2001). Studies have suggested a possible role of biomarkers as important indicators of survival and treatment decision for this patient group (Haddad and Shin 2008; Leemans et al. 2011; Peter et al. 2013).

Cytokines such as interleukin-2 (IL-2), interleukin-6 (IL-6), granulocyte–macrophage colony-stimulating factor (GM-CSF) and tumor necrosis factor-α (TNF-α) play an important role in host immune response and cancer (Feghali and Wright 1997). IL-2 and GM-CSF have impact on tumor suppression by stimulating leukocytes with antitumor activity (Rapidis and Wolf 2009). TNF-α mediates inflammation and stimulates pro-angiogenic activities essential for tumor growth (Balkwill and Mantovani 2001).

B.-Å. Andersson (✉) · S. Löfgren · N. Laytragoon-Lewin
Microbiology Laboratory, Division of Medical Services, Department of Laboratory Services, Ryhov County Hospital, 551 85 Jönköping, Sweden
e-mail: bengt-ake.andersson@lj.se

F. Lewin
Department of Oncology, Ryhov County Hospital, 551 85 Jönköping, Sweden

J. Lundgren
Department of ENT, Karolinska University Hospital, 141 86 Stockholm, Sweden

M. Nilsson
Futurum - The Academy for Healthcare, County Council, Ryhov County Hospital, 551 85 Jönköping, Sweden

L.-E. Rutqvist
Department of Scientific Affairs, Swedish Match AB, 118 85 Stockholm, Sweden

IL-6 is a pluripotent acute inflammatory cytokine mainly produced by T cells and endothelial cells (Feghali and Wright 1997). C-reactive protein (CRP) is a general acute-phase protein that increases in inflammations, infections and at tissue damages (Black et al. 2004). It has been suggested that CRP also is elevated in chronic inflammations and cancers (Heikkila et al. 2007).

In this study, a possible role of plasma IL-2, IL-6, GM-CSF, TNF-α and CRP as biomarkers for survival of HNSCC patients was investigated. Our results suggest that CRP and TNF-α might be used in combination with TNM staging for predicting survival of HNSCC patients. The use of these biomarkers in conjunction with TNM staging as a base for individualized treatment should be further investigated.

Patients and methods

Patients and healthy controls

This prospective study was performed during 2000 throw 2005 and includes 100 newly diagnosed patients that accepted to participate. Included were 27 females and 73 males, with histologically confirmed head and neck squamous cell carcinoma (HNSCC). No exclusion criteria were applied. The median age was 62 years (range 29–85). The treatment consisted of surgery and/or radiotherapy. Patients were treated and followed from the first visit up to 36 month at Department of Oncology and Department of Otolaryngology, Karolinska Hospital in Sweden. Age, sex, tumor site, metastasis, date of diagnosis, TNM stage, treatment and mortality were recorded. Heparinized blood was obtained from the patients at the time of diagnosis and before any treatment. Clinical staging was done according to TNM Classification of Malignant Tumours of the International Union Against Cancer (Sobin and Wittekind 2002).

For comparison, heparinized blood obtained from 48 healthy individuals, five female and 43 male, with a median age of 55 years (range 41–69), was included. The controls and patients were not matched and there were differences in gender ($p = 0.03$) and age ($p = 0.001$). A sample size of 45 individuals in each group was calculated to be sufficient based on alpha 0.05 with a power of 80 %.

The plasma was separated within three hours and stored at −80 °C until analyzed.

The Ethical Review Board of the Karolinska Institute approved this study and all participants gave their formal consent to participate in this study.

Plasma cytokines and CRP analysis

Levels of plasma cytokines, IL-2, IL-6, GM-CSF and TNF-α were analyzed by ELISA (Invitrogen Corporation CA). Plasma CRP levels were analyzed using Siemens Advia 1800 (Siemens Healthcare, Erlangen, Germany). All experiments were carried out according to the manufacturer's instructions.

Statistics analyzes

Data are presented as numbers, percentages, means and standard deviations. In the analysis of agreement between cytokines and CRP with tumor stages or tumor location, Spearman's rank correlation was used. Student's *t* test was used to evaluate differences in plasma cytokine and CRP levels between patients and controls. Differences in survival were analyzed with the help of Cox regression. In the Cox regression, patients were classified according to tumor stage (I–IV), combinations of cytokine and CRP levels. The tumor stages were in some analysis combined (I + II and III + IV). When analyzing 2 × 2 contingency tables, Fisher's exact test and odds ratios (including 95 % confidence intervals) were calculated. A p value ≤0.05 was considered as statistically significant.

Results

Characteristics of patients and overall survival

Characteristics of patients are shown in Table 1. There were differences in survival according to gender and TNM stage, although none of them were statistically significant.

The survival in the age group 29–55 years was significantly ($p = 0.027$) higher than the survival in age group 70–85 years.

Plasma CRP and cytokine levels

The result from analyses of CRP and cytokine levels in plasma is shown in Table 2. There was a significant difference ($p < 0.001$) between the mean value of the CRP level in patient plasma compared with the level in control plasma. The distribution of CRP values was homogeneous in the control group, whereas the patients could be divided into two clusters (Fig. 1a). Forty patients had a CRP value higher than the mean (H) and 60 a value lower than the mean (L).

The patient plasma IL-2 and GM-CSF mean values were lower than controls although not significantly. IL-6 and TNF-α in the patients were higher than plasma levels of controls. However, only the difference in the TNF-α level was significant ($p < 0.001$).

The distribution of TNF-α level in patients and controls was homogeneous (Fig. 1b). Using mean TNF-α value, the patients were divided in two groups. Forty-five patients had

Table 1 Characteristics of patients

Parameter	Number	Patient survival 36 months (%)	p value
Sex			
Male	73	38 (52)	0.12
Female	27	19 (70)	
Age (years)			
29–55	25	18 (72)	0.027[a]
56–61	24	13 (54)	
62–69	23	15 (65)	
70–85	28	11 (39)	
Tumor TNM stage			
I	4	3 (75)	
II	33	22 (67)	
III	19	9 (47)	
IV	44	23 (52)	
I + II	37	25 (67)	0.14
III + IV	63	32 (51)	
Tumor site			
Oral cavity	28	13 (46)	
Oropharynx	35	25 (71)	
Hypopharynx	7	1 (14)	
Salivary glands	6	4 (67)	
Lymph nodes, unknown primary	5	3 (60)	
Larynx	19	11 (58)	

[a] Age group 29–55 compared with age group 70–85 years

Table 2 Plasma levels of cytokines and CRP of controls ($n = 48$) and HNSCC patients ($n = 100$)

	Controls	Patients	p value
CRP (mg/L)	1.8 (1.79)[a]	7.94 (5.83)	<0.001
IL-2 (pg/mL)	4.5 (21.62)	0.37 (2.96)	0.07
IL-6 (pg/mL)	3.83 (12.80)	10.73 (50.54)	0.5
GM-CSF (pg/mL)	0.26 (1.61)	0.07 (0.47)	0.27
TNF-α (pg/mL)	20.12 (5.64)	26.51 (6.15)	<0.001

[a] Mean (SD)

a TNF-α value higher the mean (H) and 55 a value lower than the mean (L). Neither the level of CRP nor the level of TNF-α in the patient plasma did correlate with tumor location or tumor TNM stage (data not shown).

Regarding gender, 20 out of 73 male (27 %) and 4 out of 27 (15 %) female had high levels of both plasma CRP and TNF-α ($p = 0.29$). Regarding age, 10 out of 28 (36 %) patients in the age group 70–85 years and 3 out of 25 (12 %) in the age group 29–55 years had high levels of both plasma CRP and TNF-α ($p = 0.06$).

Survival of patients in relation to plasma CRP and TNF-α level

Out of the 100 patients, 57 survived and 43 died during the 36-month follow-up. The influences of CRP and TNF-α level on survival are shown in Table 3. There was a significantly ($p = 0.007$) higher surviving rate in patients with low levels of CRP compared to patients with high levels. There was a trend ($p = 0.07$) of higher survival rate in patients with low levels of TNF-α compared with high level of TNF-α. By combining a high level of CRP and TNF-α, a group with the lowest survival rate (33 %) was detected (Table 3).

Survival of patients in relation to plasma levels of CRP and TNF-α in combination with TNM stage

The number of patients and survival rates in relation to TNM stages I, II, III and IV is shown in Table 1. There was no difference in survival related to TNM stage. For further analysis, the patients were divided in two groups, TNM I + II ($n = 37$) and TNM III + IV ($n = 63$). The survival rates are shown in Table 1 and Fig. 2a. The difference was not statistically significant.

By combining TNM groups with plasma CRP and TNF-α, patients were divided into four groups (Fig. 2b). The patients designated HH had high levels of both CRP and TNF-α. The patients designated L had low levels in either or both of the biomarkers. In the group TNM I + II, eight (22 %) patients were HH and 29 (78 %) were L. In TNM III + IV, 16 (25 %) patients were HH and 47 (75 %) were

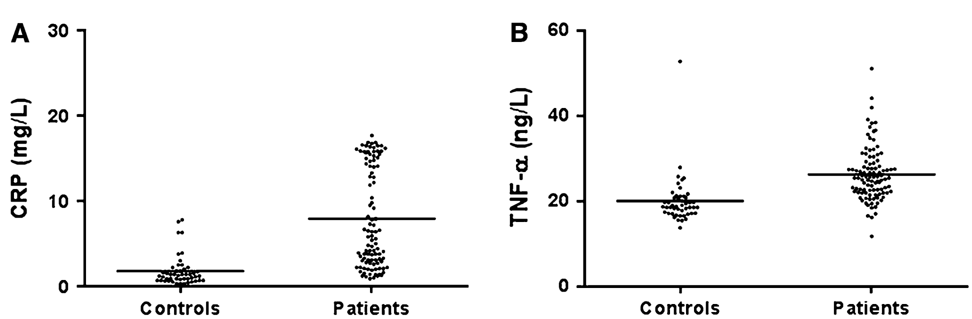

Fig. 1 Plasma levels of CRP (**a**) and TNF-α (**b**) in HNSCC patients and healthy controls

Table 3 Levels of plasma CRP and TNF-α in correlation with 36 months clinical outcome of the patient

	Number	Survival (%)	Death (%)	*p* value	OR (95 % CI)
CRP					
H	40	16 (40)	24 (60)	0.007	3.2 (1.4–7.5)
L	60	41 (68)	19 (32)		
TNF-α					
H	45	21 (47)	24 (53)	0.07	2.2 (1.0–4.8)
L	55	36 (65)	19 (35)		
CRP + TNF-α					
H/H[a]	24	8 (33)	16 (67)	0.0095	3.6 (1.4–6.6)
L[b]	76	49 (64)	27 (36)		

[a] High in both markers

[b] Low in either or both of the markers

L. Patients with stage III + IV and HH had a significantly ($p = 0.02$) lower survival rate than patients with stage III + IV and L. Also, the patients in stage I + II and HH had a lower survival rate than patients with stage I + II and L, although not statistically significant.

Discussion

In accordance with previous studies (Peter et al. 2013; van der Schroeff et al. 2012), the overall survival rate of HNSCC patients was higher in TNM group I + II than in TNM group III + IV, although not statistically significant We found that high levels of both CRP and TNF-α were significantly related to short survival time of HNSCC patients. Thus, the combination of plasma CRP and TNF-α might be a prognostic biomarker in HNSCC with a potential to differentiate the survival rate within a TNM stage.

This prospective study was performed at a county hospital laboratory with state-of-the-art methods. The treatment and the follow-up of the patients were performed at a cancer institution, which assured consistency in clinical work. No patient was lost to follow-up. The limitations are in the small number of patients, which precluded more detailed analyses according to, e.g., clinical stage, tumor site and treatment in combination with plasma level of CRP and TNF-α. It also resulted in wide confidence intervals of survival estimates and cytokine levels. There was a possible selection bias due to the lack of information on the number of patients denying to participate. Another limitation is the lack of information on HPV status. A possible influence of HPV on survival and the correlation to levels of biomarkers should be further evaluated.

HNSCC is considered to be an immune suppressing disease as illustrated by the number of T cells, T cell function and secretion of immune suppressing cytokines (Bose et al. 2008; Hoffmann et al. 2004). The lower level of plasma IL-2 and GM-CSF indicated in our study is in agreement with such an immune defect (Young 2006). Smoking has been suggested to be related to HNSCC (Lewin et al. 1998). Cigarette smoke induces massive cell death, alters gene expression (Laytragoon-Lewin et al. 2011) and influences various inflammatory proteins in blood or body fluids (Airoldi et al. 2009; Lewin et al. 1998; Malfertheiner and Schutte 2006). This local inflammatory condition might lead to tumor progression and a more compromised immune response (Balkwill and Mantovani 2001). The elevated levels of plasma CRP, IL-6 and TNF-α in our study

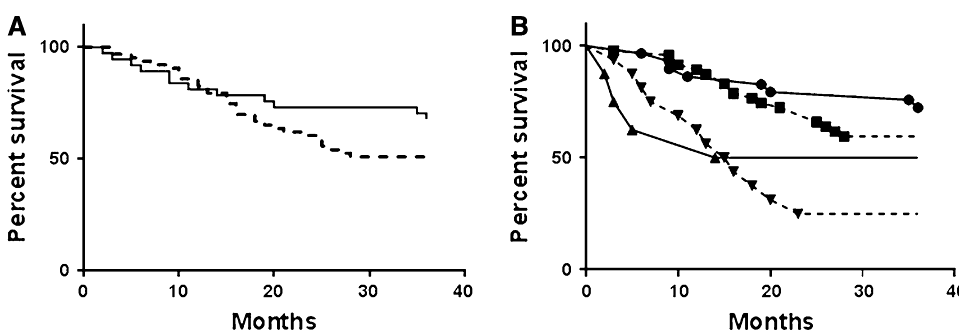

Fig. 2 Cox regression model of survival time for HNSCC patient in relation to: **a** TNM stage I + II (*solid line*) and III + IV (*dashed line*). **b** TNM stage I + II and III + IV in combination with plasma CRP and TNF-α levels. Patients with TNM stage I + II and high levels of CRP and TNF-α (*solid line with triangle*), TNM stage I + II and low levels of either CRP and TNF-α or both (*solid line with circle*), TNM stage III + IV and high levels of CRP and TNF-α (*dashed line with inverted triangle*), TNM stage III + IV and low levels of either CRP and TNF-α or both (*dashed line with square*)

might possible be due to this cancer microenvironment (Coussens and Werb 2002).

The elevated level of CRP in HNSCC patients compared to controls confirms the results of other studies (Chen et al. 1999; Peter et al. 2013). It has been reported that elevated TNF-α levels in nasopharyngeal carcinoma patients is associated with metastasis and poor prognosis (Hsiao et al. 2009; Lu et al. 2011). We have now extended the role of an elevated level of TNF-α to patients with HNSCC. By combining TNM stage and the plasma levels of CRP and TNF-α, the survival rates of HNSCC patients could be further divided. The patients with TNM stage III + IV and high levels of both CRP and TNF-α had the lowest survival rate. In contrast, patients with stage III + IV and low levels of CRP or TNF-α had no difference in survival rate compared to that of patients with stage I + II.

An influence of age on survival was indicated by the finding that the patients in the youngest age group had a significantly higher survival rate compared to the oldest. One explanation to this difference might be the trend ($p = 0.06$) to a higher level of CRP and TNF-α in the oldest group. The elderly population is usually more smokers than the younger population. Cigarette smoke might influence inflammation cytokines and CRP levels (Laytragoon-Lewin et al. 2011), and as a consequence of age and level of inflammation, differences in survival time were detected in these HNSCC patients.

Our investigation suggests that CRP and TNF-α could be used as prognostic biomarkers in combination with TNM staging for prognosis and individualized treatment of HNSCC patients. Plasma CRP and TNF-α analyses are rapid, cost effective and suitable for clinical practice. These biomarkers could identify patients with a lower risk of treatment failure despite high clinical TNM stage and patients with a higher risk of treatment failure, despite a lower clinical TNM stage.

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Conflict of interest No potential conflict of interest was disclosed.

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